AgroLedger

Supply Chain Traceability for Canadian Sheep Federation

Table of contents

01 Project Overview

02 System Architecture

03 Key Features

04 Technical Challenges & Solutions

05 Demonstration

06 Success Metrics & Results

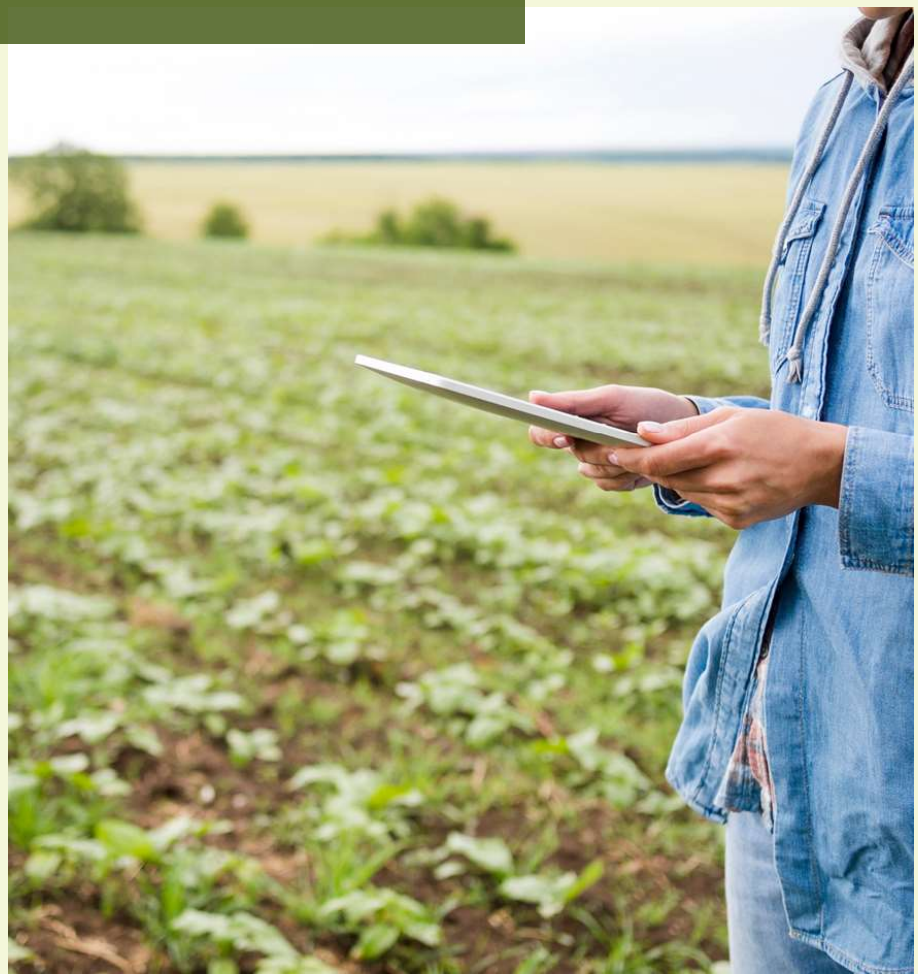
07 Future Enhancements

08 Q&A

01
Project Overview

Introduction

AgroLedger is a blockchain-based solution developed to enhance transparency, traceability, and accountability in the Canadian sheep supply chain.

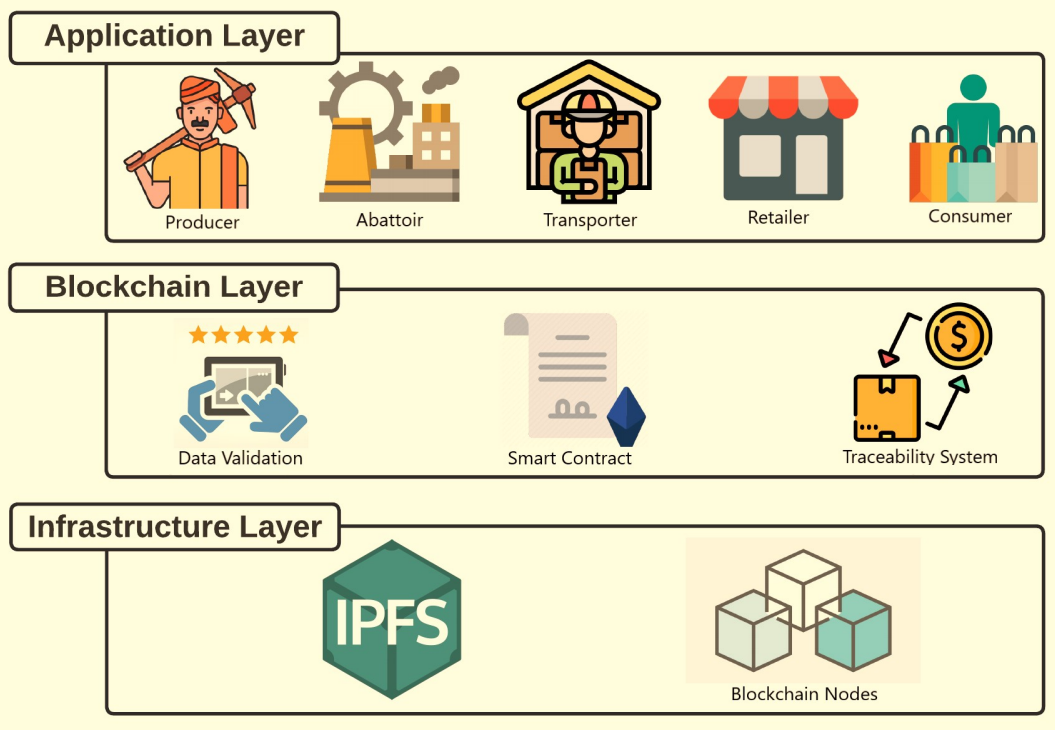


Our Objective

The system tracks key lifecycle events of sheep, including registration, movement, and slaughter, with role-based access for various stakeholders.

02. System Architecture

High Level Architecture Diagram



Architecture Components:

Blockchain:
Hyperledger Fabric for secure and immutable records.

API Layer:
Node.js backend for processing API requests and interactions with the blockchain.

Front-End:
React.js with role-based access control (RBAC).

IoT Simulation:
Simulated API calls for temperature and GPS data.

IPFS:
Decentralized file storage for certificates and documents.

03

Key Features

Key Features

Smart Contracts

Automates sheep lifecycle events such as registration, movement tracking, and slaughter logging.

IoT Data Integration

Real-time data integration of temperature and location tracking using API calls.

QR Code Integration

Each sheep has a unique QR code to provide stakeholders with real-time data access via scanning.

IPFS Integration

Decentralized storage for important documents like health certificates and movement logs.

Role-Based Access Control (RBAC)

Ensures secure, role-specific access to data.

04
Technical Challenges & Solutions

Technical Challenges & Solutions

Challenges	Solutions
• Defining complex smart contract interactions for multiple stakeholders.	• Developed smart contracts to allow easy updates and future expansions.
• Simulating IoT data without physical devices.	• Used API-based IoT simulation for real-time data interaction.
• Managing user roles, certificates, and secure access.	• Automated user management with Hyperledger Fabric CA and JWT-based API authentication.
• Storing documents on a decentralized network with IPFS.	• Integrated IPFS with the blockchain to store and retrieve records securely.

05

Live Demonstration

Walkthrough the System

Step 1

Registering a sheep through the system interface (producer role).

Step 2

Tracking the sheep's movement from farm to slaughterhouse.

Step 3

Accessing decentralized records stored on IPFS (via blockchain).

Step 4

Simulated IoT data entry (temperature and location updates).

Step 5

Scanning the QR code to access sheep data (consumer role).

06
Success Metrics & Results

Project Successes

Traceability

Full traceability from sheep registration to slaughter and end product delivery.

Security

Secure and immutable records stored on the blockchain.

IoT Integration

IoT integration provides real-time tracking and monitoring capabilities.

QR code Integration

QR code integration allows seamless data access for stakeholders.

IPFS

IPFS ensures secure, decentralized storage for critical records.

07
Future Enhancements

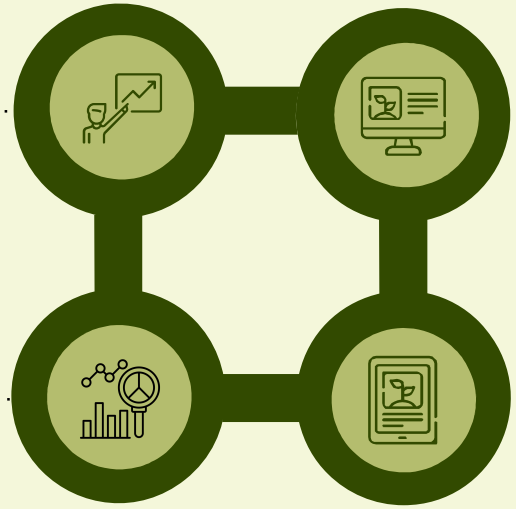
Future Enhancements

Scalability

Expand the system to include other livestock types and agricultural sectors.

Predictive Analytics

Use analytics to monitor livestock health and predict performance trends.



Physical IoT Integration

Replace simulated data with actual IoT devices for real-time tracking.

Enhanced IPFS Use

Store additional records, such as health certificates, and allow broader decentralized access.

08

Questions & Answers

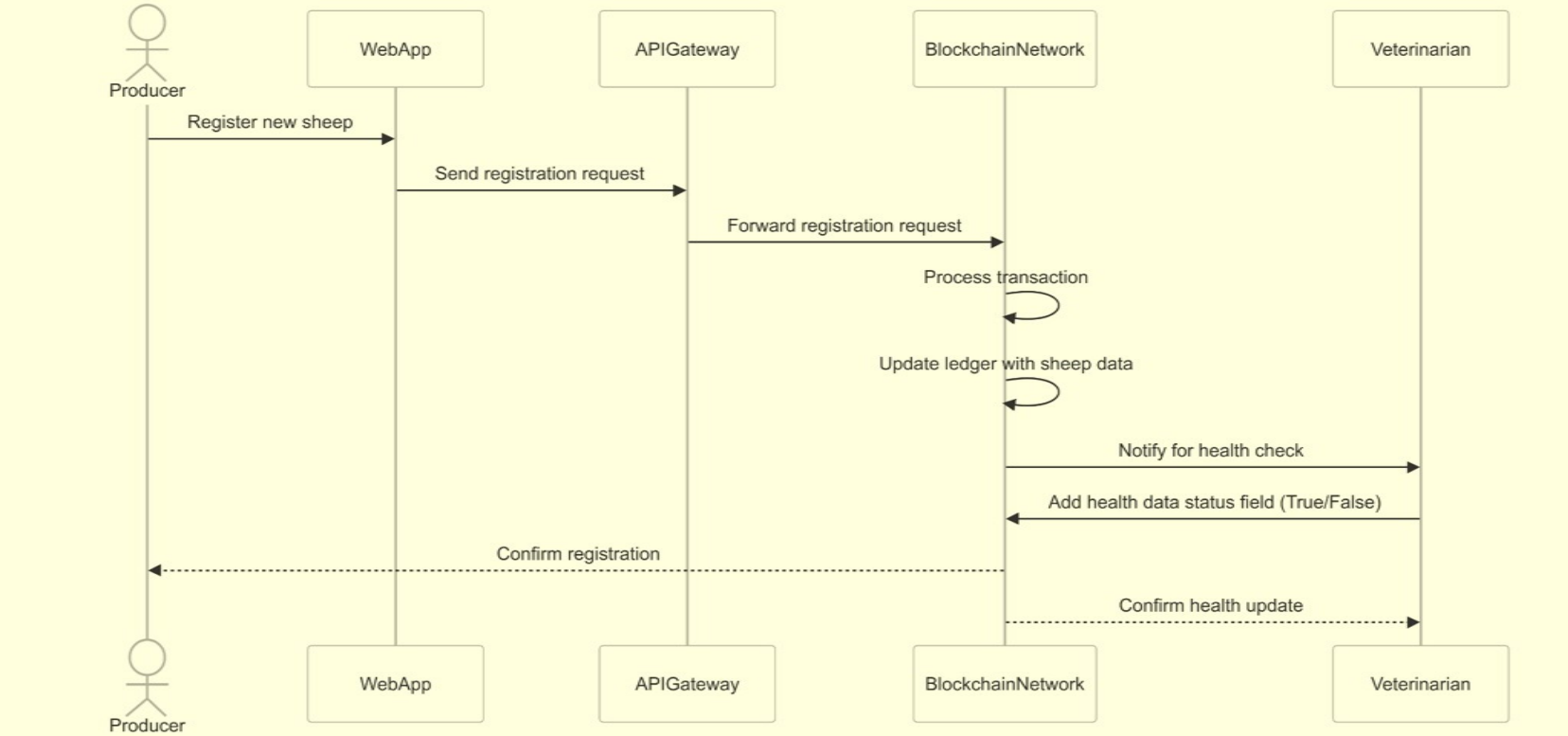
Open the floor for any questions

Thanks!

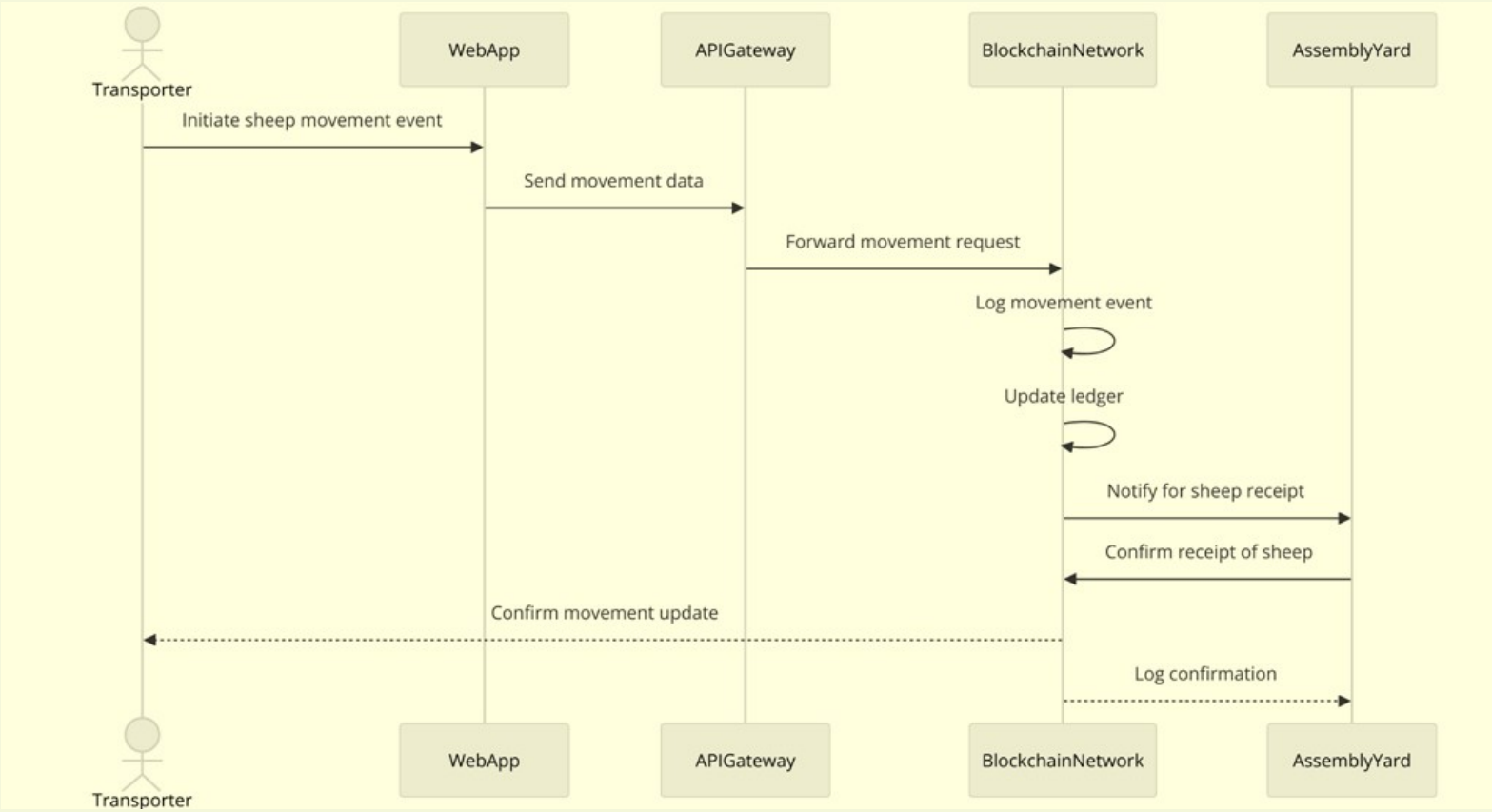
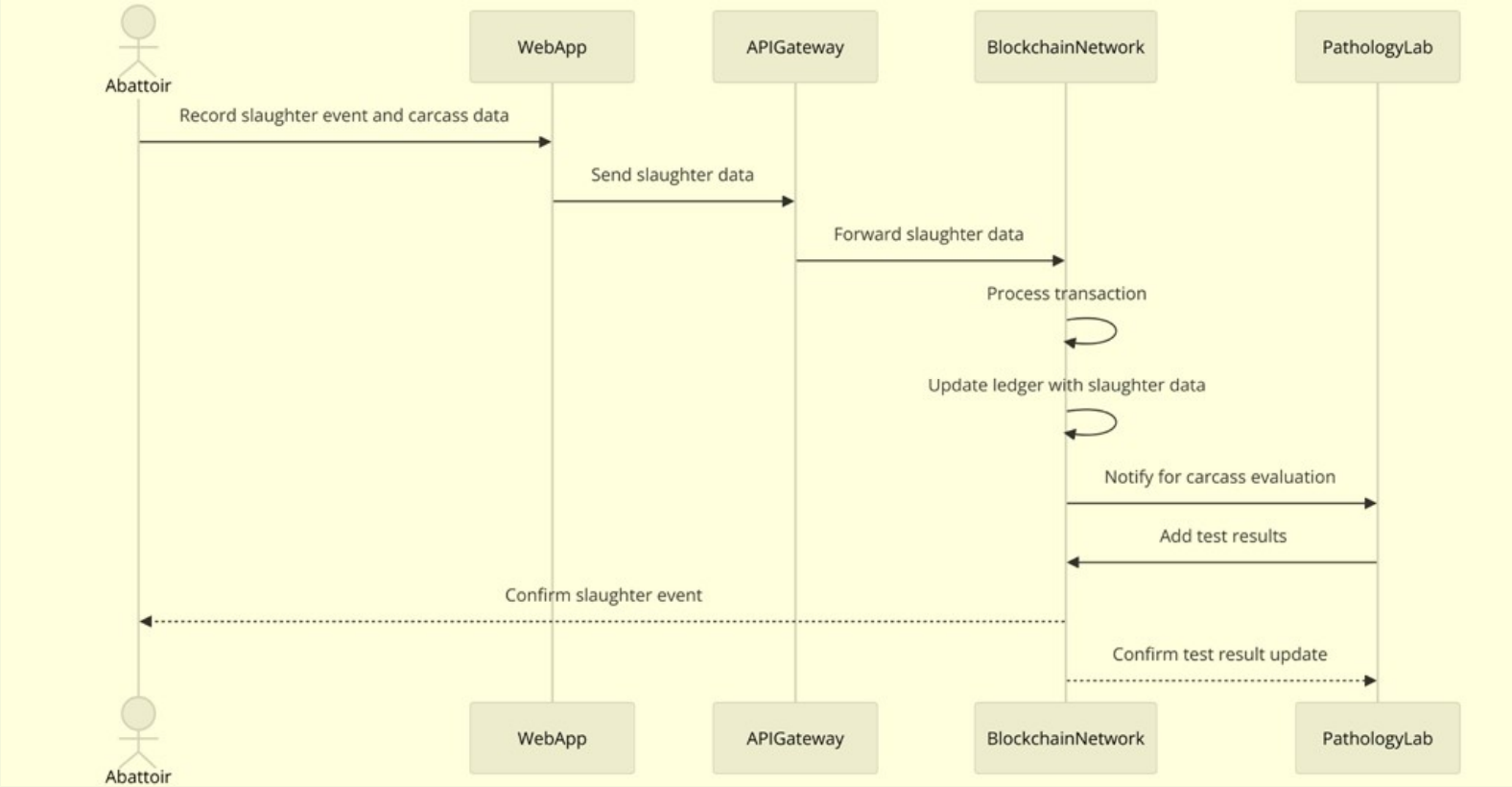
Team Members:

Joseph Varghese
Chirag Vaghasiya
Yash Patel
Rahul Kacha
Tri Thanh Alan Inder Kumar

Sheep Registration Sequence Diagram



Sheep Movement Tracking Diagram



Slaughter and Carcass Disposal Diagram